▫ 03 Probability models – the binomial distribution

Consider each of the scenarios below.

Problem 1
If a fair coin (heads and tails are equally likely) is flipped 20 times what is the probability that exactly 10 heads will be observed?

Problem 2
8% of items produced in a factory are defective. A potential customer is considering purchasing a batch of such items and will buy the batch provided that at most 2 items in a randomly selected sample of 12 are defective. How likely is it that the batch will be purchased?

Problem 3
A large group of students have a gender split of 60% male and 40% female. If a committee of 5 is randomly selected what are the chances that there will be at least two females on the committee?

While the context of each problem differs, there are some common elements to all three.

In all cases the problem can be viewed as a series of <u>trials</u> with each trial capable of ending in two possible outcomes (nominally labelled success or failure).

The actual outcome of each trial is unknown in advance of the trial, but the probability of each outcome is the same for all trials so that the trials (in this respect) are identical.

For example, in the coin flipping problem a single flipping of a coin constitutes a trial with the outcomes being heads (with probability 50%) and tails (with probability 50%).

In the defective items problem, the customer will test each of the 12 items in the sample (i.e., 12 trials). Each test may result in the item being found to be defective (with probability 8%) or non-defective (with probability 92%).

In the committee selection problem each of the 5 committee's members are observed in turn (5 trials) are classified as male (with probability 60%) and female (with probability 40%).

In general if we let X = the number of successes out of a total of n trials where each the probability of success in each trial is p then we say X has a <u>binomial probability</u> distribution with parameters n and p.
i.e., $X \sim \text{bin}(n, p)$

We can calculate the probability of k successes in n trials using the <u>probability mass function</u>

$Pr(X = k) = {}^nC_k\, p^k\, (1-p)^{n-k}$ where $k = 0, 1, 2, \ldots n$. That is a whole number between 0 and n. And $0 \leq p \leq 1$.

We can use counting principles, and the idea of independent events from probability, to see why this mass function is correct.

<u>Problem 1</u>

What is the probability that exactly 10 heads will be observed?

Note that although 10 heads in 20 flips is what you would expect from a fair coin and is a more likely outcome that any other (e.g. 9 heads, 11 heads etc.) it is still relatively unlikely outcome. Less than 1 in 5 such experiment of flipping 20 coins will result in exactly 10 heads and 10 tails.

Problem 2
How likely is it that the batch will be purchased?

Problem 3
What are the chances that there will be at least two females on the committee?

Exercise
How likely is it that there will be less than 3 males on the committee of 5 described in the last problem?

Alternatively, we could have tackled the problem by assigning a new variable Y.
i.e. Let Y = # males on a committee of 5. Y~ bin(5, 0.6). $\Pr(Y = k) = {}^5C_k\, 0.6^k\, 0.4^{5-k}$

We require $\Pr(Y \geq 3)$. Exercise Complete the problem and show that the again you get 68%.

Binomial distribution in R

Suppose we have X ~ bin(8, 0.4)

To obtain Pr(X = 3).
R> dbinom(3, 8, 0.4)
[1] 0.2786918

To obtain $\Pr(X \leq 3)$:
R> pbinom(3, 8, 0.4)
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[1] 0.5940864
```

To obtain $\Pr(X \geq 3)$:
R> 1 - pbinom(2, 8, 0.4)
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[1] 0.6846054
```

Alternatively use
R> pbinom(2, 8, 0.4, lower = FALSE)

To obtain and plot Pr(X = k) for k = 0,1, ... 8
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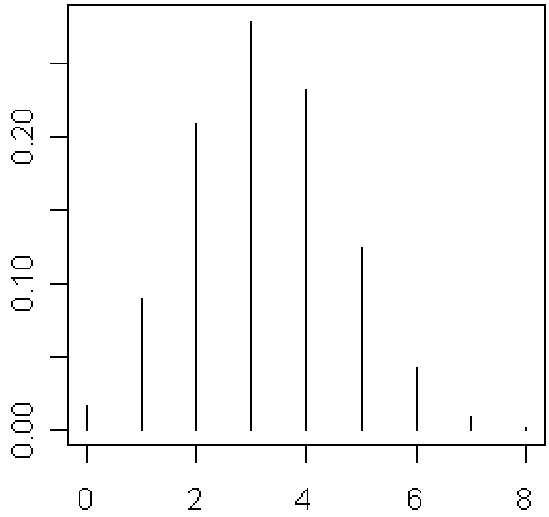
R>plot(0:8, dbinom(0:8, 8,0.4), type= "h")
```



Output from
plot(0:8, dbinom(0:8, 8,0.4), type= "h")

Exercise
Write down the R command which will produce a plot similar to the one above which shows the likelihood of each possible number of tails that might result when a fair coin is flipped 20 times.

Exercise

A survey of all first years, who have a laptop computer, reveal that 120 have PCs (windows), 50 have Macs and 30 have machines running Linux.

(a) If a sample of 6 first years are randomly selected, how likely it is that at most 2 will have Macs?
(b) If a sample of 8 are selected, how likely it is that at least 4 but less than 7 will have PCS?
(c) If a sample of 9 are selected, how likely it is that more than 7 will **not** have Linux?

Use R to confirm your answers